



Southwest Houston 345 kV Reliability Project (24RPG032)

High Voltage Planning

CenterPoint Energy Houston Electric



Project Background

- CenterPoint Energy transmission system serves over 21 GW of load in the greater Houston metropolitan area.
- Latest 345 kV tie-lines were added in 2007 (Hillje to W.A. Parish double circuit) and 2018 (Gibbons Creek to Zenith double circuit).
- Since 2018, areas along the Hillje to W.A. Parish corridor in Fort Bend and Wharton Counties have seen a high level of renewable generation siting close to the 345 kV transmission lines.

Project Background

- Table below shows the generation capacity connected to 345 kV in Southwest Houston.

Generation Type	EOY2018 (MW)	EOY2023 (MW)	EOY2025 (MW)
Nuclear	2573	2573	2573
Conventional	2993	2993	2993
Solar	0	2807	4660
Storage	0	150	1323
Wind	0	151	599
Total	5566	8674	12,148

Project Background

- In 2022, total annual congestion on 345 kV W.A.Parish-Obrien was \$48 million¹.
- In 2023, total annual congestion on 345 kV STP-W.A.Parish ckt 39 was \$124 million².
- ERCOT noted 345 kV STP-W.A.Parish ckt 39 as one of the highest congested elements in the 2022 Regional Transmission Plan (RTP).

¹ <https://www.ercot.com/files/docs/2023/01/26/December-2022-ERCOT-Operations-Report-Public.docx>

² <https://www.ercot.com/files/docs/2024/01/26/december-2023-ercot-operations-report-public.docx>

Project Background

- ERCOT 2023 RTP study identified two reliability projects within CEHE footprint.

RPG Project Number	Operations Line Name	Branch	Line Length (Miles)	Ownership	Existing Ratings (MVA) NMRL/EMGY/LD SD/COND/RELAY	Year Needed
2023-C1	OB_WAP98	44000-44500 <CKT 98> W_A_P_POI5A TO OBRIEN_B345A	16.98	CNP	1450.0/1450.0/1450.0/1450.0/3129.0	2025
	OB_WAP99	44000-44500 <CKT 99> W_A_P_POI5A TO OBRIEN_B345A	16.98		1450.0/1450.0/1450.0/1450.0/3129.0	2025
2023-C4	STPWAP39	5915-44000 <CKT 39> SO_TEX_345A TO W_A_P_POI5A	68.29	CNP/AEP	1137.0/1195.0/1195.0/1195.0/2311.0	2025

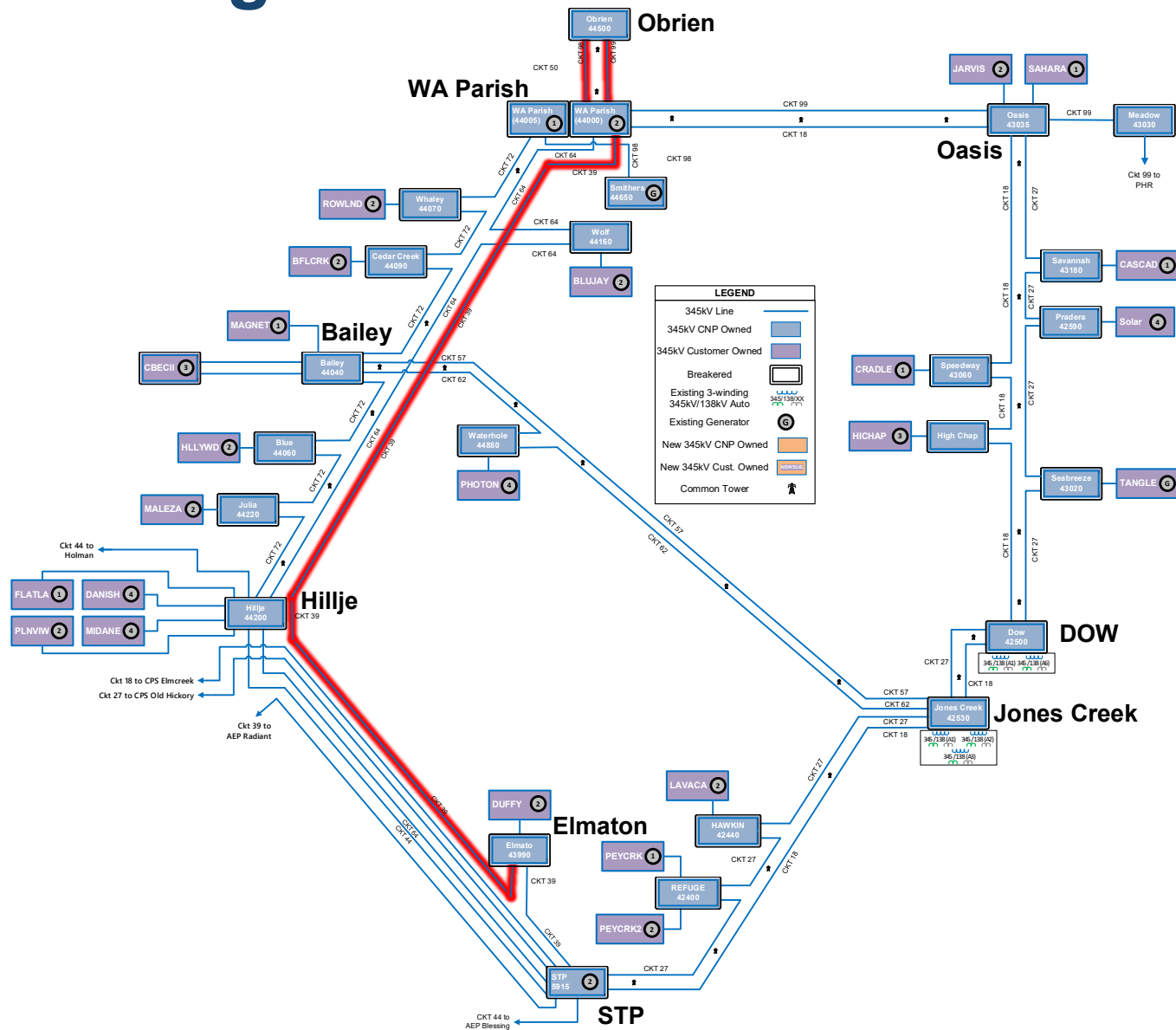
- The 345 kV W.A.Parish-Obrien ckt 98 & ckt 99 double circuit line was constructed in 1963 using lattice towers with 2-795 ACSR conductor, and majority of the structures are approximately 50+ years old.
- The 345 kV STP-W.A.Parish ckt 39 line was constructed in 1974/1977 using lattice towers. The line is comprised of multiple segments, with different owners. The entire circuit is constructed using 2-795ACSR conductor.

Study Background

- The circuits recommended in 2023 RTP analysis were seen to have thermal overload concerns on multiple planning events as shown below.

Branch Name	Rating	CEHE Planning Events under which the Lines are Overloaded							
	NMRL/EMGY/LDSD/COND/RELAY	P0	P1	P2	P5	P7	P3	P6x	P6o
5915-43990 <CKT 39> SO_TEX__345A TO ELMATO_POI_5	1137.0/1450.0/1587.0/1450.0/3441.0	No	No	No	No	No	No	No	No
43990-44000 <CKT 39> ELMATO_POI_5 TO W_A_P__POI5A	1137.0/1195.0/1195.0/1195.0/2311.0	No	Yes	Yes	No	Yes	Yes	Yes	No
44000-44500 <CKT 98> W_A_P__POI5A TO OBRIEN_B345A	1450.0/1450.0/1450.0/1450.0/3129.0	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No
44000-44500 <CKT 99> W_A_P__POI5A TO OBRIEN_B345A	1450.0/1450.0/1450.0/1450.0/3129.0	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No

Pre-Project Configuration



Study Analysis

- CenterPoint Energy studied 20+ options to resolve the thermal overload concerns seen in the 2023 RTP analysis.
- CenterPoint Energy recommends to proceed with Option 2cd as the most cost-effective solution to resolve the thermal loading concerns seen in 2023 RTP analysis.

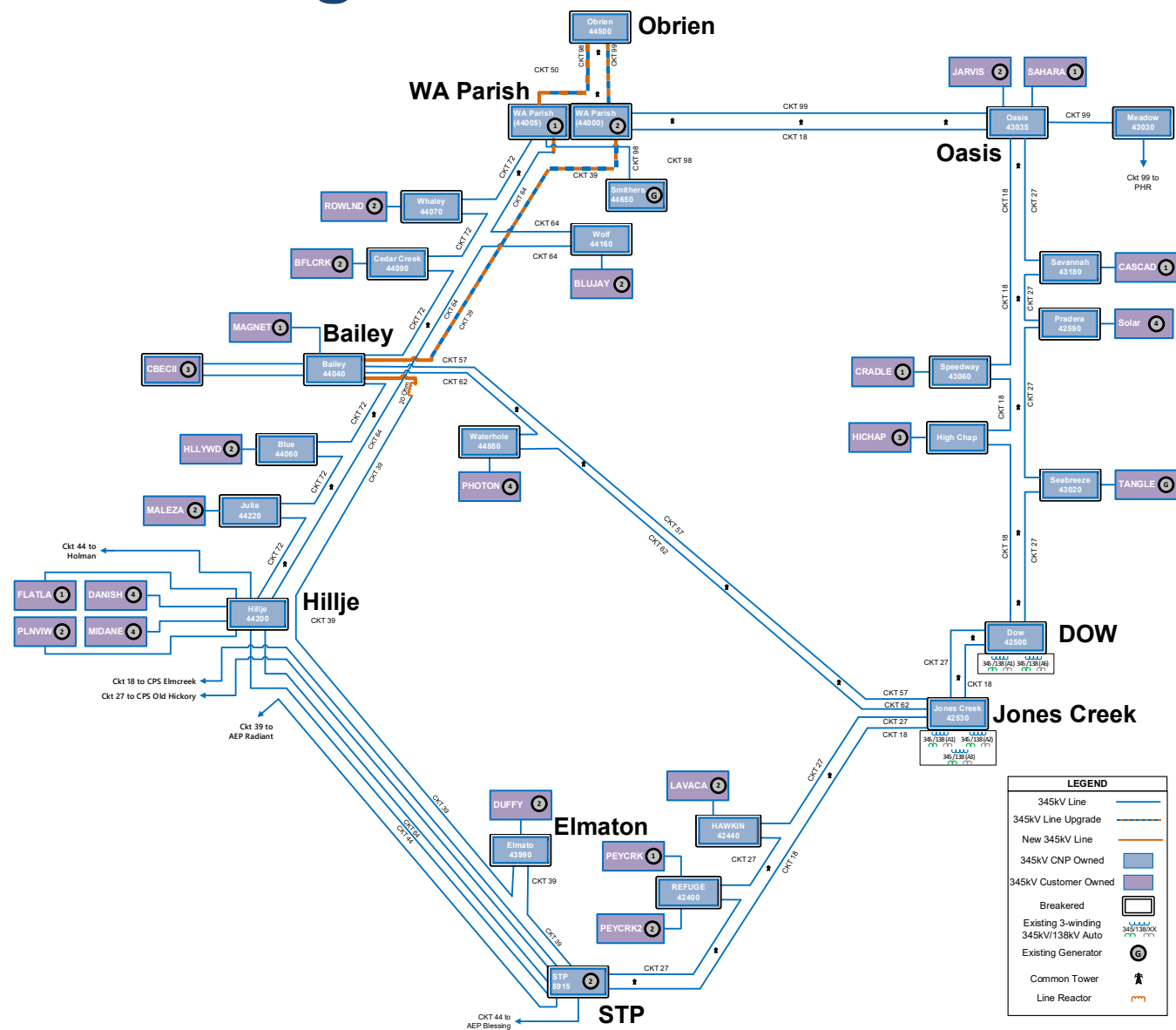
Project Overview

- Loop 345 kV Elmaton-W.A.Parish ckt 39 into 345 kV Bailey substation.
- Install 20 Ohm line reactor at Bailey on 345 kV Elmaton-Bailey ckt 39.
- Reconductor/rebuild approximately 31 miles of 345 kV Bailey-W.A.Parish ckt 39 w/ 3-959 ACSS to minimum Normal/Emergency ratings of 2987/3147 MVA.
- Reconductor/rebuild approximately 17 Miles of 345 kV W.A.Parish-Obrien ckt 98 and ckt 99 w/ 3-959 ACSS to minimum Normal/Emergency ratings of 2987/3147 MVA.
- Relocate 345 kV W.A.Parish-Wolf ckt 64 from 345 kV W.A.Parish North bus to 345 kV W.A.Parish South bus.
- Relocate 345 kV W.A.Parish-Obrien ckt 98 from 345 kV W.A.Parish North bus to 345 kV W.A.Parish South bus.

Project Overview Continued

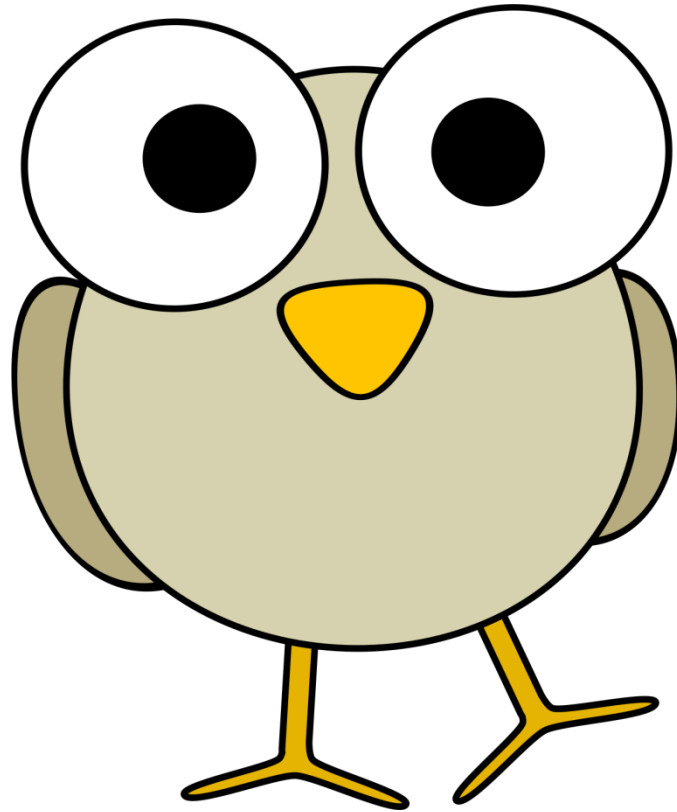
- Upgrade substation equipment on 345 kV W.A.Parish-Jeanetta ckt 72 to minimum Normal/Emergency ratings of 2038/2645 MVA.
- Upgrade substation equipment on 345 kV W.A.Parish-Whaley ckt 72 to minimum Normal/Emergency ratings of 2710/2812 MVA.
- Upgrade substation equipment at 345 kV Bailey substation to a 63 kA fault interrupting capability.
- Upgrade substation equipment at 345 kV Obrien substation to a 63 kA fault interrupting capability.
- **Total cost of the proposed RPG Project - \$569.322 million**
 - The above cost includes temporary construction cost to minimize outages during construction time.

Option 2cd Configuration



Project Timeline

- Anticipating no CCN applications/submittals.
- Construction expected to begin in 2025.
- Expected in-service date of before summer peak 2028.
- Proposed RPG submittal comment period ended on November 4, 2024.
 - ERCOT independent Review to be complete by April 3, 2025.



??